Toward Sustainable AI Infrastructure: Reinforcement Learning for Energy-Aware Data Center Scheduling and Climate-Conscious Computing

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1 MOTIVATION & PROBLEM

AI's exponential growth is consuming unprecedented energy. Global data centers account for an estimated 1-2% of worldwide electricity, a figure set to rise sharply as AI workloads intensify.

Most schedulers use simple heuristics designed for throughput alone, ignoring electricity price volatility, renewable availability, and thermal efficiency.

In the Global South, this problem is acute: unreliable grids, constrained capacity, and ambitious AI adoption goals collide. Intelligent, energy-aware scheduling is therefore not merely a technical optimization, it is a prerequisite for equitable, sustainable digital development.

2 RESEARCH QUESTIONS

Can deep RL outperform heuristic schedulers on energy cost?
Comparing PPO agent vs. Least-Loaded, Round-Robin, and Random baselines across 30 evaluation episodes.

Can LSTM networks predict server workload reliably?
Using network-traffic features to forecast CPU load, feeding the RL state space.

Does this generalise to resource-constrained contexts?
Evaluating lightweight RL scheduling for sustainable AI in Global South infrastructure.

3 SYSTEM ARCHITECTURE

Workload	Energy	PPO Agent	Outcome
Network traffic features	Electricity demand signals	Schedules GPU jobs	Reduced energy cost
LSTM	Chronos	RL	SLA ✓

4 RL REWARD MECHANISMS

Energy cost penalty
Reward = −(server load × job size × electricity price), incentivising off-peak and low-load assignment.

Sleep-state bonus
Idle servers (load < 5%) receive a positive reward for entering sleep, reducing idle power draw.

Price-aware deferral
Jobs deferred during 3× peak tariff periods earn a reward bonus; unnecessary deferrals incur a penalty.

SLA violation penalty
Server overload (> 85% utilisation) triggers a −2 penalty, enforcing service quality constraints.

-51.1%	~0	p<0.05
ENERGY COST	SLA VIOLATIONS	SIGNIFICANCE
vs. best heuristic	vs. 160+ for heuristics	All Welch t-tests

5 ENERGY CONSUMPTION BY POLICY

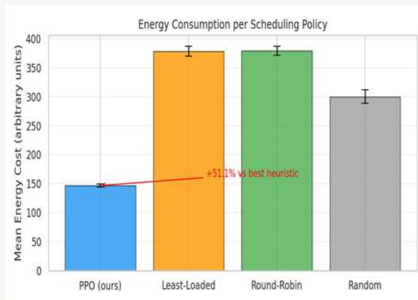

Fig. 1 — Mean energy cost (± std) per scheduling policy over 30 episodes. PPO achieves 51.1% reduction vs. the best heuristic.

6 COMPARATIVE EVALUATION — 30 EPISODES

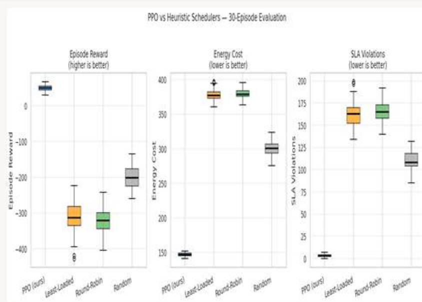

Fig. 2 — Distribution of episode reward, energy cost, and SLA violations across 30 episodes. PPO dominates all three metrics with low variance.

7 SUMMARY RESULTS

Policy	Reward	Energy Cost	SLA Violations
PPO (ours) ★	~+30	~150	~0
Least-Loaded	~−320	~380	~163
Round-Robin	~−330	~380	~165
Random	~−200	~300	~112

8 LSTM WORKLOAD FORECASTING

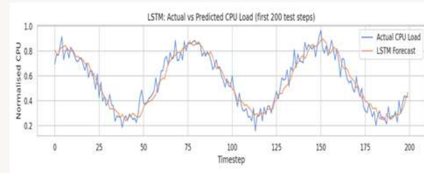

Fig. 3 — LSTM forecast vs. actual normalised CPU load (first 200 test steps). Model captures daily cyclical patterns with low lag.

A 2-layer LSTM (64 units, 50-step look-back, dropout 0.2) trained on server network-traffic features predicts CPU utilisation 1 step ahead.

9 LSTM TRAINING CONVERGENCE

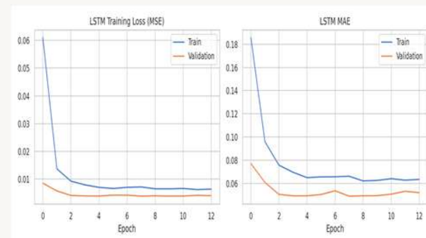

Fig. 4 — Training/validation MSE & MAE over epochs. Converges within 4 epochs; validation MSE < 0.006, no overfitting observed.

10 METHODS SUMMARY

Component	Method
Workload forecast	2-layer LSTM, 64 units
Energy forecast	Chronos-Bolt (48M, zero-shot)
RL scheduler	PPO (Stable-Baselines3)
Policy network	MLP [128,128], γ = 0.99
Training	150,000 steps, 4 parallel envs
Evaluation	30 episodes, Welch t-test

11 BROADER IMPLICATIONS

Climate Resilience
Energy-aware scheduling reduces AI's carbon footprint, directly supporting climate goals.

Global South
RL scheduling helps balance constrained grid capacity with growing AI service demand.

Responsible AI
Applying AI to optimise AI infrastructure is a concrete path to systemic sustainability.

Digital Equity
Efficient infrastructure lowers costs, making AI services more accessible in emerging economies.

12 CONCLUSIONS & FUTURE WORK

A PPO reinforcement learning agent achieves 51.1% lower energy cost and near-zero SLA violations compared to conventional scheduling heuristics, all improvements are statistically significant.

The framework integrates LSTM workload forecasting and Chronos zero-shot energy demand forecasting as state features.

Future directions:
- Deployment on real GPU clusters with live electricity pricing
- Multi-objective optimisation (cost, carbon intensity, latency)
- Adaptation to renewable energy constrained infrastructure representative of Global South contexts

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